



**INSTITUTE OF
BUSINESS MANAGEMENT**
BOARD OF ADVANCED STUDIES & RESEARCH



APPLIED BUSINESS ANALYTICS

FINAL REPORT



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1. Selection & Segregation Criteria (About Data)

We have analyzed 24 Fast-Moving Consumer Goods (FMCG) companies which are listed under the Food and Personal Care Products category of the Pakistan Stock Exchange (PSX). The choice was carefully selected to give a representative sample of the industry, with large multinationals (MNCs), traditional local businesses with decades of history, and fast-growing new players. Our research area, due to the given sector, will include the peculiarities of the dividend distribution in one of the most stable and consumer-oriented areas of the Pak economy.

The main aim of this sample was to test the dividend reliability within a conclusive six-year range between the years 2020 and 2025. This time period is one that is especially important because it compelled businesses to walk the fine line of economic polarities. It started with the most factual operational lockdowns of the world pandemic (2020-2021) followed by a phase of historic currency devaluations and hyper-inflationary pressure (2022-2024) and ended by a phase of high interest rates and market stabilization in 2025. We might use the dividend announcements made during these tumultuous years as a objective gauge of which companies had the financial strength to pay its shareholders during such hard times of extreme margin squeezes.

The distinction between regular and irregular payers was clearly distinguished by our analysis of the consistent market leaders and the irregular payers. The dividend payers with the same consistency became those which had strong fundamentals, i.e., strong free cash flow and disciplined corporate governance culture. The latter firms usually exercised their pricing power to direct the costs to the consumer, making shareholder rewards top of the agenda. On the other hand, non-paying or irregular firms marked the structural issues of the time, including excessive leverage, reliance on imported raw materials and hard trade-off between business reinvestment and instant investor dividends.1. At-Tahur Ltd. (**PREMA**)

2. Barkat Frisian Agro (**BFAGRO**)
3. Big Bird Foods Ltd. (**BBFL**)
4. Bunnys Limited (**BNL**)
5. Clover Pakistan (**CLOV**)
6. Colgate Palmolive Pakistan (**COLG**)
7. Fauji Foods Ltd (**FFL**)
8. FrieslandCampina Engro Pakistan (**FCEPL**)
9. Gillette Pakistan (**GLPL**)

10. Ismail Industries Ltd (**ISIL**)
11. Matco Foods Ltd (**MFL**)
12. Mitchell's Fruit Farms Ltd (**MFFL**)
13. Murree Brewery Co. Ltd (**MUREB**)
14. National Foods Ltd (**NATF**)
15. Nestlé Pakistan Ltd (**NESTLE**)
16. Quice Food Industries Ltd (**QUICE**)
17. Rafhan Maize Products Co. Ltd (**RMPL**)
18. Shezan International Ltd (**SHEZ**)
19. Shield Corporation Ltd (**SCL**)
20. The Organic Meat Company Ltd (**TOMCL**)
21. Treet Corporation Ltd (**TREET**)
22. Unilever Pakistan Foods Ltd (**UPFL**)
23. Unity Foods Ltd (**UNITY**)
24. ZIL Limited (**ZIL**)
25. Al-Shaheer Corporation Ltd (**ASC**)

additional selection procedure was well structured to target the companies that showed a firm dedication to rewarding their shareholders with dividends. It is true that most of the companies in the industry have had issues like declining income, unstable economy, and inconsistent consumer demand but not many of them have recorded a steady stream of paying dividends over the years. In 2019-2025, this uniformity started to form the major point of difference and underline the financial firmness and shareholder-oriented approaches of some companies. Upon careful reading, it was seen that very few of the listed companies have paid dividends on a regular basis during this time. These are organizations that are not just profitable, but also committed towards maintaining investor confidence. The highest issuers of dividends are:

1. Colgate-Palmolive Pakistan (**COLG**)

2. FrieslandCampina (**FCEPL**)
3. Ismail Industries (**ISIL**)
4. Murree Brewery (**MUREB**)
5. National Foods (**NATF**)
6. Nestlé Pakistan (**NESTLE**)
7. Rafhan Maize (**RMPL**)
8. Shezan International (**SHEZ**)
9. Unilever Pakistan Foods (**UPFL**)
10. ZIL Limited (**ZIL**)

Most of the sample companies have exhibited a much less predictable trend of dividend payments as compared to a few companies who have been found to pay their shareholders consistently. Most of these organizations paid dividends infrequently, and irregularly, with some having intervening periods and different amounts paid out, or entirely stopped paying dividends after 2020. Others either chose to suspend dividends due to the decreasing profitability, the increasing cost of doing business or because of the necessity of investing the profits back into business so as to remain profitable. Then there are the ones that have merely not paid out any dividends over the past few years, indicating either a fiscal constraint, or a decision to pursue other business growth at the expense of dividends. This anomaly reflects both the difficulty of smaller or less established players in having a stable dividend policy, and the difference between strong market leaders and those unable to strike a balance between stability and investor expectations.

But further subdivided it into three groups viz. first Dividend Leaders (2020-2025) Secondly Solid but Moderate Payers and thirdly Irregular / No Dividends.

#	Category	Symbol	Company	Dividend Frequency	Notes
1	Leaders	NESTLE	Nestle Pakistan	Quarterly	Extremely high payouts, never missed

2	Leaders	UPFL	Unilever Pakistan Foods	Quarterly	Strong growth, consistent every year
3	Leaders	RMPL	Rafhan Maize	Quarterly	High-value dividends, very steady
4	Leaders	MUREB	Murree Brewery	Quarterly	Reliable, strong cash returns
5	Leaders	NATF	National Foods	Quarterly	Consistent, moderate growth
6	Solid but Moderate	COLG	Colgate-Palmolive Pakistan	Semi-annual	Reliable, steady
7	Solid but Moderate	FCEPL	FrieslandCampina	Annual	Smaller but consistent
8	Solid but Moderate	ISIL	Ismail Industries	Annual	Regular but modest
9	Solid but Moderate	SHEZ	Shezan International	Annual	Smaller payouts, but consistent
10	Solid but Moderate	ZIL	ZIL Limited	Annual	Steady since 2021
11	Irregular/No Dividends	ASC	Al-Shaheer Corp	None	No regular dividends
12	Irregular/No Dividends	PREMA	At-Tahur Ltd	None	No regular dividends
13	Irregular/No Dividends	BFAGRO	Barkat Frisian Agro	None	No regular dividends

14	Irregular/No Dividends	BBFL	Big Bird Foods Ltd	None	No regular dividends
15	Irregular/No Dividends	BNL	Bunnys Limited	None	No regular dividends
16	Irregular/No Dividends	CLOV	Clover Pakistan	None	No regular dividends
17	Irregular/No Dividends	FFL	Fauji Foods Ltd	None	No regular dividends
18	Irregular/No Dividends	MFFL	Mitchell's Fruit Farms	None	No regular dividends
19	Irregular/No Dividends	QUICE	Quice Food	None	No regular dividends
20	Irregular/No Dividends	SCL	Shield Corp	None	No regular dividends
21	Irregular/No Dividends	TOMCL	The Organic Meat	None	No regular dividends
22	Irregular/No Dividends	TREET	Treet Corp	None	No regular dividends
23	Irregular/No Dividends	UNITY	Unity Foods Ltd	None	No regular dividends

The companies which have been featured in this analysis were well chosen based on their sound dividend policies and the performance of their annual dividend per share. This was not just a certain process of selecting firms which made regular dividend announcements, but it was a process of selecting those which invested in supporting their shareholders over the long run. We studied the dividend announcements in 2020 to 2025 and could differentiate the organizations that continued with their high and consistent dividends and those that failed or ceased their payment.

The remarkable fact is that the annual dividends per share of the highlighted companies represent financial power as well as restrained corporate management. Even during economic uncertainties these firms have been able to earn enough earnings and cash flows to enable them to make regular distributions. The fact that they are capable of sustaining or increasing dividends over the years, is indicative of resilience, stability, and client-focused shareholder policy.

Unlike other companies who paid intermittent dividends, or even stopped paying dividends, the chosen companies are the actual dividend leaders in the Food and Personal Care Industry of Pakistan Stock Exchange. Selected & Approved Companies

Category	Symbols	Company	Frequency	Notes
Leaders	NESTLE	Nestle Pakistan	Quarterly	Extremely high interim & final payouts
Leaders	UPFL	Unilever Pakistan Foods	Quarterly	Very strong growth, multiple large interim dividends
Leaders	RMPL	Rafhan Maize	Quarterly	Regular payouts
Leaders	MUREB	Murree Brewery	Quarterly	Reliable, strong cash returns
Leaders	NATF	National Foods	Quarterly	Consistent payouts
Solid but Moderate	COLG	Colgate-Palmolive Pakistan	Semi-annual	Reliable, steady

2. Selected and Approved Companies

Colgate-Palmolive (Pakistan) Limited (COLG).

Colgate-Palmolive Pakistan is an institution of the FMCG industry in the nation. When the company was incorporated in 1977 as National Detergents Limited, it experienced a transformational change in 1990 when the company entered into a strategic alliance with the international Colgate-Palmolive Company (USA). This partnership enabled the company to

integrate global research and development and local production dexterity. Based in Karachi and listed in the PSX, the company has been dominated by the performance compared to its rivals to become a market leader in oral care and household hygiene.

Its brand portfolio is varied and has deep penetration which is a strength of the company. Although Colgate is a household name in Pakistan when it comes to toothpaste, the company also enjoys a huge market share of the fabric care and dish washing divisions. It has a strategic distribution of its manufacturing bases in Karachi, Kotri, and Lahore and this guarantees it an efficient distribution channel that covers the furthest regions of the nation. In addition to business, the firm is known to have its Bright Smiles, Bright Future program that has taught oral hygiene to millions of Pakistani children.

- Brands: Colgate Toothpaste and Brushes, Palmolive Soap and Shampoo, Brite, Softlan and Lemon Max.
- Strategic Advantage: Privy to international proprietary preparations and the national distribution system is solid.
- Production Centres: Three technologically advanced plants with high efficiency and quality assurance.

Murree Brewery Company Limited (MUREB).

Murree Brewery is one of the oldest public limited companies in the region as it was founded in 1860. It started as a British personnel institution to serve the needs of the British people in the colonialism period and has since become a very diversified industrialized group. Although business operations are in a challenging regulatory environment, the Murree Brewery has been able to achieve a strong presence in the Pakistan Stock Exchange by using its heritage and brand equity to preserve a distinctive market niche.

The operations of the company are divided into three main segments, which include Liquor, Glass, and Tops Food and Beverages. The company has taken an aggressive move to diversify its non-alcoholic brands to appeal to the wider population of Muslims in the world despite its alcoholic products forming a substantial part of its heritage. Tops brand is a familiar household brand of fruit juices, squashes and pickles and their mineral water and carbonated soft drinks are still riding. Moreover, this self-sufficiency is supported by the fact that they have an in-house glass manufacturing unit where they can make quality containers to their own products and also their external users.

- Diversified Portfolio: Beers and spirits, Tops brand juices and sparkling water (named “Murree Sparkletts”) are in its portfolio.
- Industrial Integration: Has a specialized glass division so as to have self-sufficient packaging.

- History: It is a historic institution in the corporate history of Pakistan, having been operating more than 160 years.

National Foods Limited (NATF)

NFL is a pioneer of the packaged food industry in Pakistani kitchen which has been credited to being modernized by National Foods Limited. NFL was founded in 1970 and specialized in hygiene and convenience and it changed the consumption of traditional spices by shifting the market to loose and unbranded spices to standardised and packaged recipe mixes. It is now a multi-category food giant that has a vision of becoming a global food company with its exports being currently reached in over 40 countries spread over five continents.

The secret of the success of the company lies in its capability to combine the traditional desi flavor with the contemporary food processing technology. It has one of the most modern Faisalabad plants in the region that can deal with huge amounts of spices, salts and sauces. NFL has a wide portfolio with the range of basic ingredients as well as ready to eat meals to accommodate the busy lifestyle of the urban consumer. They have created a brand that people are proud of as a country by paying attention to Corporate Social Responsibility (CSR) and sustainable sourcing.

- Product Categories: Spices, recipe mixes, pickles, ketchups, jam, and desserts.
- Global Reach: Good penetration in the Middle East, UK and North America serving the Pakistani diaspora.
- Innovation: A pioneer in the convenience foods, which saves a lot of time to prepare meals at home.

Nestle Pakistan Limited (NESTLE).

The Nestle Pakistan is a subsidiary of the Swiss multinational Nestle S.A. which is arguably the most powerful force in the nutrition and dairy industry of the country. It has been transformational in the Pakistani agricultural economy since its founding in 1979 (then as Milkpak Limited). The company has one of the biggest milk collection systems in the world, which cooperates with hundreds of thousands of local farmers, which has the effect of enhancing the quality and quantity of dairy.

Nestle Pakistan is listed on the PSX and is concentrated on the Nutrition, Health and Wellness. It has a wide product portfolio that includes infant food, bottled water, dairy, and confection. Such brands as Nescafe and Milkpak have become leaders in the category. The company has large production plants in Sheikhpura and Kabirwala that are the best standards of food safety and environmental sustainability in Pakistan. They are committed towards their creation of Shared

Value (CSV), thus their growth will add value to the well being of the rural communities where they conduct their operations.

- Key Brands: Nestle Milkpak, Everyday, Nescafe, Maggi and Nestle Pure Life.
- Agricultural Impact: Water stewardship: Intensive dairy farming support programs.
- Market Position: The gold standard in the local food industry of the quality assurance and nutritional science.

Rafhan Maize Products Company Limited (RMPL).

Rafhan Maize Products is an important connection point between the agricultural sector of Pakistan and the industrial environment. Rafhan is the leading maize processor in the country, which was founded in the early nineties and is mostly owned by Ingredion Incorporation. Rafhan is not like other FMCG companies that only deal with retail stores, it is a vital supplier of dozens of other industries such as textiles, pharmaceuticals, paper, and confectioneries.

The industrial power of the company is in the possibility to use maize (corn) to produce high value products, liquid glucose, starches, and corn oil. Rafhan is also a household name though it has a consumer division that manufactures popular dessert mixes such as custards, jellies, and cornflour. This two-fold business strategy the balance between the B2B industrial supply and B2C consumer products offers the firm with a diversified and stable flow of revenues. It has a history of making regular dividends and this has ensured that it has become a favorite stock among long term investors in the Pakistan Stock Exchange.

- Industrial Powerhouse: Is the vendor of critical raw materials to the textile and pharmaceutical industries.
- Consumer Favorites: Market leader in dessert brands (Rafhan Custard, Jelly, Cornflour).
- Agro-Processing Excellence: It is the use of technology to maximize the value-add of the domestic maize crops.

Unilever Pakistan Foods Limited (UPFL)

Unilever Pakistan Foods Limited (UPFL) is the specific food unit of the operation of Unilever in Pakistan. Although Unilever has been established in the region since 1948, UPFL is an independent listed company that deals in the high growth packaged food segment. The company uses Unilever international knowledge in the field of food science and sustainability to provide products that are adapted to the local taste.

The UPFL buildings are established on the basis of Power Brands which have enormous loyalty. Knorr has reinvented a noodles and soup category in Pakistan, and Rafhan (through Unilever umbrella under the specific category of food brands) would be the brand to use with desserts.

The Hellmann's brand also leads the market in the category of sauces and dressings with its company. The strategy of UPF has been driven on the concept of purpose led growth with an aim of enhancing the nutritional value of its products and lowering its environmental footprint by use of sustainable plastic packaging and responsible sourcing.

- Portfolio Core: Knorr (noodles/bouillons), Rafhan (desserts), Glaxose-D and Energile.
- Strategic Insight: Adapts global brands to local tastes (e.g. Knorr Chatpatta noodles) successfully.
- Productivity: The advantages of the global supply chain and digital marketing of Unilever.

3. Data Marts and Key Performance Indicators (KPIs)

In this project, the financial and performance data are arranged in thematic Data Marts, where a data mart is a significant business area. Each data mart contains Key Performance Indicators (KPIs) based on measuring performance, efficiency, profitability, liquidity and market valuation. With the support of this structure, it is possible to make decisions on reporting, dashboarding, and analytics. Approved KPS's for Project by our respected faculty Sir Nasir Iqbal

Operating Performance Data Mart

KPIs:

- Sales
- Gross Profit
- Operating Profit
- Profit Before Tax
- Profit After Tax
- Earnings Per Share (EPS)
- Gross Profit Margin
- Operating Profit Margin
- Net Profit Margin

Dividends & Payouts Data Mart

KPIs:

- Dividend Per Share (Total)
- Dividend Payout Ratio
- Price Earnings Ratio (P/E)
- Market Price Per Share
- Net Assets Per Share

Balance Sheet Data Mart

KPIs:

- Current Assets
- Non-current assets
- Current liabilities
- Non-current liabilities
- Net Assets

Each Data Mart will have a column of Symbol, Company name, Year wise, quarter wise

Category	Metric	Unit
Operating Performance		
	Sales	PKR Million
	Gross Profit	PKR Million
	Operating Profit	PKR Million
	Profit Before Tax	PKR Million
	Profit After Tax	PKR Million

	Gross Profit Margin	%
	Operating Profit Margin	%
	Earnings Per Share (EPS)	PKR
Dividends & Payouts		
	Dividend Per Share (Total)	PKR
	Dividend Payout Ratio	%
	Price Earning Ratio (P/E)	-
	Market Price Per Share	PKR
	Net Assets Per Share	PKR
Balance Sheet		
	Current asset	PKR Million
	Non-current Assets	PKR Million
	Current Liabilities	PKR Million
	Non-current liabilities	PKR Million

	Net Assets	PKR Million
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4. Table Designs (OLTP AND OLAP)

OLTP Table Design

Column Name	Data Type	Description
company_symbol (PK)	VARCHAR(10)	PSX ticker symbol
company_name	VARCHAR(100)	Company name
sector	VARCHAR(50)	Industry / sector
fiscal_year	INT	Financial year
period_type	VARCHAR(10)	Annual / Quarterly
quarter	VARCHAR(2)	Q1–Q4 (nullable)
sales	DECIMAL(18,2)	Total sales
gross_profit	DECIMAL(18,2)	Gross profit
operating_profit	DECIMAL(18,2)	Operating profit
profit_before_tax	DECIMAL(18,2)	Profit before tax
profit_after_tax	DECIMAL(18,2)	Profit after tax

eps	DECIMAL(10,2)	Earnings per share
gross_profit_margin	DECIMAL(5,2)	Gross profit margin (%)
operating_profit_margin	DECIMAL(5,2)	Operating profit margin (%)
net_profit_margin	DECIMAL(5,2)	Net profit margin (%)
dividend_per_share	DECIMAL(10,2)	Dividend per share
dividend_payout_ratio	DECIMAL(5,2)	Dividend payout ratio (%)
market_price_per_share	DECIMAL(10,2)	Market price per share
pe_ratio	DECIMAL(10,2)	Price–Earnings ratio
net_assets_per_share	DECIMAL(10,2)	Net assets per share
current_assets	DECIMAL(18,2)	Current assets
non_current_assets	DECIMAL(18,2)	Non-current assets
current_liabilities	DECIMAL(18,2)	Current liabilities
non_current_liabilities	DECIMAL(18,2)	Non-current liabilities
net_assets	DECIMAL(18,2)	Net assets / Equity

OLAP table design

Financial Performance Mart

Column Name	Data Type	Description
symbol	VARCHAR(10)	Company ticker
company_name	VARCHAR(100)	Name of company
year	INT	Financial year
quarter	VARCHAR(2)	Q1–Q4
sales	DECIMAL(18,2)	Total sales
gross_profit	DECIMAL(18,2)	Gross profit
operating_profit	DECIMAL(18,2)	Operating profit
pat	DECIMAL(18,2)	Profit after tax
gp_margin	DECIMAL(5,2)	Gross profit margin (%)
op_margin	DECIMAL(5,2)	Operating profit margin (%)
eps	DECIMAL(10,2)	Earnings per share

Dividend Payout Mart

Column Name	Data Type	Description
-------------	-----------	-------------

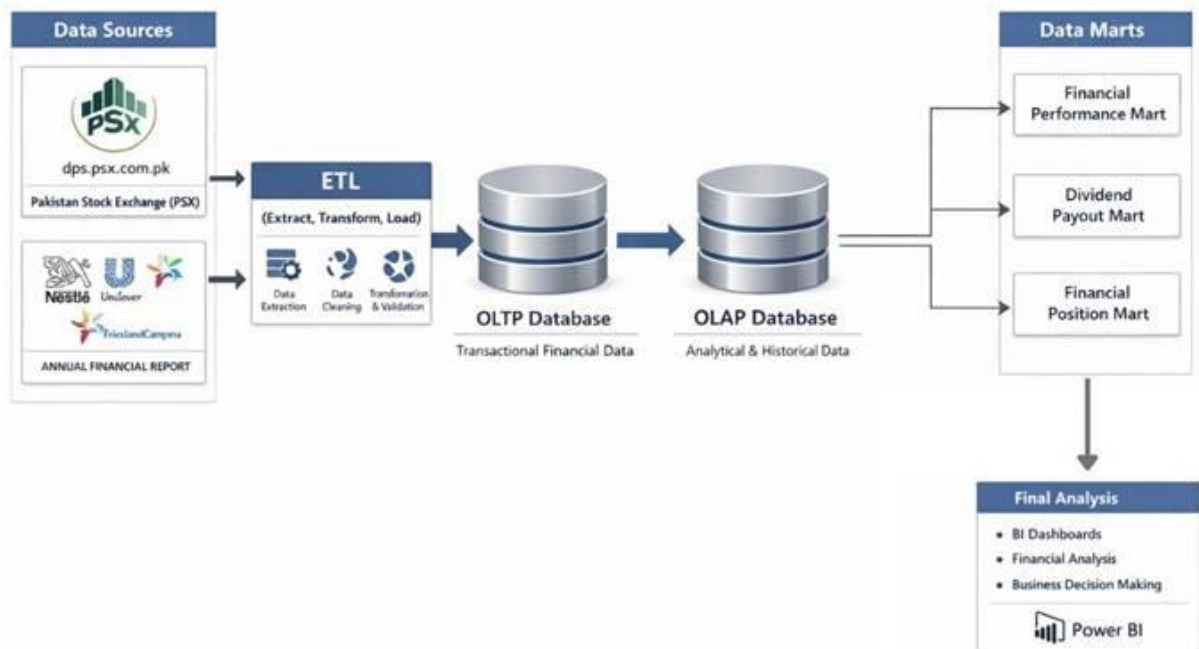
symbol	VARCHAR(10)	Company ticker
company_name	VARCHAR(100)	Name of company
year	INT	Financial year
quarter	VARCHAR(2)	Q1–Q4
dps	DECIMAL(10,2)	Dividend per share
payout_ratio	DECIMAL(5,2)	Dividend payout ratio (%)
pe_ratio	DECIMAL(10,2)	Price–Earnings ratio
market_price	DECIMAL(10,2)	Market price per share
net_assets_per_share	DECIMAL(10,2)	Net assets per share

Financial Position Mart

Column Name	Data Type	Description
symbol	VARCHAR(10)	Company ticker
company_name	VARCHAR(100)	Name of company
year	INT	Financial year
quarter	VARCHAR(2)	Q1–Q4

current_assets	DECIMAL(18,2)	Current assets
non_current_assets	DECIMAL(18,2)	Non-current assets
current_liabilities	DECIMAL(18,2)	Current liabilities
non_current_liabilities	DECIMAL(18,2)	Non-current liabilities
net_assets	DECIMAL(18,2)	Net assets / equity

Data Flow Architecture



5. PHASE – 1 (ETL)

Data Extraction

The most technical process involved the extraction step, which is mainly because the machine-readable quarterly data is not standardized throughout the Pakistan Stock Exchange (PSX).

Technical Problems and Limitations of the Platform.

- Although the original objective was to achieve complete automation with the help of web scraping and AI, multiple bottlenecks required the change of approach:
- Artificial Intelligence Extraction Breakdowns: the standard AI systems were unable to browse and properly extract structured financial tables out of complex PSX URLs.
- Data Integrity Problems: Early testing using the Gemini platform has shown that when multi-page financial statements are processed it is giving hallucinated or incomplete figures.
- pdf restrictions PDF deepseek had file-size restrictions; combining a couple of years of annual reports was to truncate data, omit values, and duplicate rows.
- Infrastructure Gaps: Following the schedule of the 2025 project and the necessity to be extremely accurate, a tailor-made web-scraping solution was postponed in favor of a proven manual-digital hybrid solution.

The Hybrid Approach

We used a semi-manual AI-assisted method of extractions to ensure a verified dataset. Information on finances was harvested by page out of official PDF annual reports. This was secure to ensure that all dividend announcements and EPS (Earnings Per Share) values were supplemented by the report of the chairman and audited financial notes.

Data Transformation

The transformation step was the quality control component, where the raw, noisy data was cleaned and normalized to be used in longitudinal analysis.

Determined Data Quality Problems.

- Errors in Formulas: The discrepancies in dividend payout ratios in which stock dividends (bonus shares) were calculated in error compared with cash dividends.
- Syntax Noise: This exists when non-numeric characters are present in numeric columns (e.g., "PKR" or "million" or "nil") and this causes the inability to perform mathematical operations on the data.
- Structural Redundancy: The creation of ghost columns and blank rows in the earliest AI-assisted OCR operation.

Corrective Actions Taken

- Reason Re-Checking: Re-calculated all the payout ratios based on the standardized formula.
- Data Scrubbing: use of Regex (Regular Expression) data cleaning to eliminate unwanted text and symbols to financial fields.
- Schema Standardization: Created a Unified naming system (e.g.: renamed Total Div and Dps to single header DividendPerShare), and then converted all them to a single unit (Millions of PKR).
- Temporal Alignment: Unified all the end-of-fiscal and year-end dates with the aim of providing an accurate year-over-year comparison of 2020 and 2025.

Data Loading

The last stage was to convert the purified data to a high-performance platform of the multi-level querying and visualization.

- Export Protocol: The data was moved into a CSV (Comma Separated Values) format which acted as the mediator between the transformation and loading layers.
- SQL Integration: Microsoft SQL Server was used to import the file to the Flat File Import wizard.
- Schema Mapping: Import Import data types were strictly defined:
 - 18 and 2 digits to represent the financial values in order to avoid rounding off.
 - Varchar of company identifiers.
 - Int for fiscal years.

Analytical Readiness: The ability to store the data in the form of a relational database allowed us to use Window Functions and Aggregates to find the trends of dividends in the 6 FMCG firms at a single run.

6. PHASE – 2 (OLTP & OLAP)

Overview

The given stage of the project is devoted to changing FMCG financial data stored in an OLTP system into an organized OLAP environment and the creation of analysis reports with the help of Microsoft Power BI. The main aim of this phase is to facilitate the efficient financial analysis through designing subject-oriented data marts and adopting financial KPIs and measures that will assist in making strategic decisions.

- The activities to be done during this stage include:
- OLAP data mart design in SQL server.
- Using financial aggregation rationale.

- Time standardization of quarterly analysis.
- Technological measures on the concept of analytical implementation with the help of DAX.
- Building financial reporting interactive power bi dashboards.

OLTP Source System Characterization.

The OLTP database FMCGFINOLTP stores the source data, and it has only one table that is transactional, financialsfmcg.

- The major OLTP Data Characteristics.
- Granular financial records on a record level.
- Frequency of quarterly reporting.
- Includes income statement, balance sheet, and dividend related measures.
- Tuned to store and be consistent as opposed to analytics.

These features make direct analytical reporting on the OLTP system ineffective and not suitable in terms of business intelligence utilization.

OLAP Database Design

In order to host analytical loads, OLAP database named FMCGFIN_OLAP has been established. The OLAP database is denormalized and based on the star-schema, each data mart illustrates a particular financial topic area.

- Design Principles Applied
- Data is organized in subject form.
- Quarterly aggregation.
- Segregation of the analysis and the transaction workloads.
- Stable financial computing logic.
- Numerical precision (2 decimal places) that is reporting ready.

Financial Performance Mart

Purpose

Financial Performance Mart is created to estimate the performance of the income statement of the FMCG firms quarterly. It allows the assessment of the increase of revenue, profitability, and efficiency of operations.

Table Creation Query

“

```

CREATE TABLE FMCG_FIN_OLAP.dbo.FinancialPerformanceMart (
    symbol VARCHAR(10),
    company_name VARCHAR(100),
    year INT,
    quarter VARCHAR(2),
    sales DECIMAL(18,2),
    gross_profit DECIMAL(18,2),
    operating_profit DECIMAL(18,2),
    pat DECIMAL(18,2),
    gp_margin DECIMAL(5,2),
    op_margin DECIMAL(5,2),
    eps DECIMAL(10,2)
);

```

Data Insertion Query

```

INSERT INTO FMCG_FIN_OLAP.dbo.FinancialPerformanceMart
SELECT
    Symbol,
    Company_Name,
    YEAR(Period) AS year,
    Fiscal_Quarter AS quarter,
    ROUND(SUM(Sales), 2) AS sales,
    ROUND(SUM(Gross_Profit), 2) AS gross_profit,
    ROUND(SUM(Operating_Profit), 2) AS operating_profit,
    ROUND(SUM(Profit_After_Tax), 2) AS pat,
    ROUND(AVG(Gross_Profit_Margin), 2) AS gp_margin,
    ROUND(AVG(Operating_Profit_Margin), 2) AS op_margin,
    ROUND(AVG(Earnings_Per_Share_EPS), 2) AS eps

```

FROM FMCG_FIN_OLTP.dbo.financials_fmccg

GROUP BY

Symbol,

Company_Name,

YEAR(Period),

Fiscal_Quarter;

“

Query Explanation

- Sales and profits are two examples of financial values that are summed up by means of SUM.
- To avoid any misrepresentation, margins and EPS are averaged.
- Period column is used to get the year.
- Fiscal quarters are normalised to Q1-Q4.
- Any numeric values are rounded off to two decimal places.

Output

	symbol	company_name	year	quarter	sales	gross_profit	operating_profit	pat	gp_margin	op_margin	eps
1	GOLG	Colgate-Palmolive (Pakistan) Limited	2020	Q2	12265.00	3729.00	2017.00	1333.00	0.30	0.16	5.49
2	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q1	14072.00	3860.00	2134.00	1488.00	0.27	0.15	6.13
3	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q2	14687.00	3929.00	2144.00	1447.00	0.27	0.15	5.96
4	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q3	12913.00	3924.00	2216.00	1558.00	0.30	0.17	6.42
5	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q4	13086.00	3462.00	1063.00	1345.00	0.26	0.08	5.54
6	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q1	19507.00	4756.00	2617.00	1718.00	0.24	0.13	7.07
7	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q2	22438.00	6142.00	3638.00	2499.00	0.27	0.16	10.30
8	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q3	16226.00	4105.00	2291.00	1698.00	0.25	0.14	7.00
9	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q4	17346.00	4135.00	1326.00	1238.00	0.24	0.08	5.10
10	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q1	27683.00	8387.00	6048.00	3701.00	0.30	0.22	15.24
11	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q2	27388.00	8047.00	5890.00	3671.00	0.29	0.22	15.12
12	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q3	24197.00	7260.00	4532.00	2976.00	0.30	0.19	12.26
13	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q4	25318.00	7919.00	4681.00	3217.00	0.31	0.18	13.25
14	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q1	30003.00	10802.00	6770.00	5104.00	0.36	0.23	21.02
15	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q2	27916.00	9806.00	7031.00	4592.00	0.35	0.25	18.92
16	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q3	28954.00	9845.00	7541.00	4727.00	0.34	0.26	19.47
17	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q4	29206.00	10576.00	6626.00	5194.00	0.36	0.23	21.39
18	GOLG	Colgate-Palmolive (Pakistan) Limited	2025	Q3	29058.00	10241.00	7090.00	4407.00	0.35	0.24	18.15
19	MUREB	Murree Brewery Company Limited	2020	Q2	2193.00	595.22	268.56	218.24	0.27	0.12	7.89
20	MUREB	Murree Brewery Company Limited	2021	Q1	3712.00	945.19	524.86	437.05	0.25	0.14	15.80
21	MUREB	Murree Brewery Company Limited	2021	Q2	2969.00	863.40	447.32	362.92	0.29	0.15	13.12
22	MUREB	Murree Brewery Company Limited	2021	Q3	2990.00	888.82	505.78	379.40	0.30	0.17	13.71
23	MUREB	Murree Brewery Company Limited	2021	Q4	3417.00	643.31	156.66	336.61	0.19	0.05	12.17
24	MUREB	Murree Brewery Company Limited	2022	Q1	3194.00	794.74	271.21	258.27	0.25	0.08	9.34
25	MUREB	Murree Brewery Company Limited	2022	Q2	5041.00	754.53	268.19	275.36	0.15	0.05	9.95
26	MUREB	Murree Brewery Company Limited	2022	Q3	3911.00	1092.00	575.35	454.33	0.28	0.15	16.42
27	MUREB	Murree Brewery Company Limited	2022	Q4	4642.00	621.89	149.09	39.81	0.13	0.03	1.44
28	MUREB	Murree Brewery Company Limited	2023	Q1	5017.00	1419.00	859.05	590.69	0.28	0.17	21.35
29	MUREB	Murree Brewery Company Limited	2023	Q2	5743.00	1237.00	722.52	561.67	0.22	0.13	20.30
30	MUREB	Murree Brewery Company Limited	2023	Q3	4923.00	1141.00	605.33	439.05	0.23	0.12	15.87
31	MUREB	Murree Brewery Company Limited	2023	Q4	5433.00	780.83	502.19	301.02	0.14	0.09	10.88
32	MUREB	Murree Brewery Company Limited	2024	Q1	7104.00	1918.00	1198.00	913.86	0.27	0.17	33.03
33	MUREB	Murree Brewery Company Limited	2024	Q2	7202.00	1762.00	1222.00	822.18	0.24	0.17	32.58

Dividend Payout Mart

Purpose

The Dividend Payout Mart emphasizes on the analysis of the shareholder returns, and thus is able to evaluate the dividend distribution policies, and the consistency of capital returns.

Market price and P/E ratio were eliminated as the means of sustaining analytical relevance and making it not market-driven volatility.

Table Creation Query

```
“  
  
CREATE TABLE FMCG_FIN_OLAP.dbo.DividendPayoutMart (  
  
    symbol VARCHAR(10),  
  
    company_name VARCHAR(100),  
  
    year INT,  
  
    quarter VARCHAR(2),  
  
    dps DECIMAL(10,2),  
  
    payout_ratio DECIMAL(5,2),  
  
    net_assets_per_share DECIMAL(10,2)  
  
);  
“
```

Data Insertion Query

```
“  
  
INSERT INTO FMCG_FIN_OLAP.dbo.DividendPayoutMart  
  
SELECT  
  
    Symbol,  
  
    Company_Name,  
  
    YEAR(Period) AS year,  
  
    Fiscal_Quarter AS quarter,  
  
    ROUND(AVG(Dividend_Per_Share_Total), 2) AS dps,  
  
    ROUND(AVG(Dividend_Payout_Ratio), 2) AS payout_ratio,  
  
    ROUND(AVG(Net_Assets_Per_Share), 2) AS net_assets_per_share
```

FROM FMCG_FIN_OLTP.dbo.financials_fmcs

GROUP BY

Symbol,

Company_Name,

YEAR(Period),

Fiscal_Quarter;

“

Query Explanation

- Measures of dividends are averaged in order to ensure uniformity.
- Facilitates the comparison between dividend policies between companies.
- Facilitates shareholder value analysis in the long run.

Output

	symbol	company_name	year	quarter	dps	payout_ratio	net_assets_per_share
1	GOLG	Colgate-Palmolive (Pakistan) Limited	2020	Q2	6.52	1.19	76.92
2	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q1	0.00	0.00	82.31
3	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q2	7.35	1.23	88.27
4	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q3	0.00	0.00	76.81
5	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q4	6.26	1.13	82.43
6	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q1	0.00	0.00	91.56
7	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q2	6.90	0.67	101.85
8	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q3	0.00	0.00	87.91
9	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q4	8.40	1.65	92.88
10	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q1	0.00	0.00	115.41
11	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q2	22...	1.49	125.53
12	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q3	0.00	0.00	107.22
13	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q4	9.05	0.68	100.16
14	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q1	0.00	0.00	130.24
15	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q2	32...	1.69	149.15
16	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q3	0.00	0.00	122.50
17	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q4	34...	1.61	143.71
18	GOLG	Colgate-Palmolive (Pakistan) Limited	2025	Q3	0.00	0.00	135.31
19	MUR...	Murree Brewery Company Limited	2020	Q2	10...	1.27	371.77
20	MUR...	Murree Brewery Company Limited	2021	Q1	5.00	0.32	421.73
21	MUR...	Murree Brewery Company Limited	2021	Q2	10...	0.76	429.85
22	MUR...	Murree Brewery Company Limited	2021	Q3	5.00	0.36	375.48
23	MUR...	Murree Brewery Company Limited	2021	Q4	10...	0.82	415.93
24	MUR...	Murree Brewery Company Limited	2022	Q1	0.00	0.00	440.39

Financial Position Mart

Purpose

The Financial Position Mart is a balance sheet format of the FMCG companies and allows the analysis of liquidity, solvency, and capital structure.

Table Creation Query

```
“  
  
CREATE TABLE FMCG_FIN_OLAP.dbo.FinancialPositionMart (  
    symbol VARCHAR(10),  
    company_name VARCHAR(100),  
    year INT,  
    quarter VARCHAR(2),  
    current_assets DECIMAL(18,2),  
    non_current_assets DECIMAL(18,2),  
    current_liabilities DECIMAL(18,2),  
    non_current_liabilities DECIMAL(18,2),  
    net_assets DECIMAL(18,2)  
);  
“
```

Data Insertion Query

```
“  
  
INSERT INTO FMCG_FIN_OLAP.dbo.FinancialPositionMart  
SELECT  
    Symbol,  
    Company_Name,  
    YEAR(Period) AS year,  
    Fiscal_Quarter AS quarter,  
    ROUND(SUM(Current_Assets), 2) AS current_assets,  
    ROUND(SUM(Non_Current_Assets), 2) AS non_current_assets,  
    ROUND(SUM(Current_Liabilities), 2) AS current_liabilities,  
    ROUND(SUM(Non_Current_Liabilities), 2) AS non_current_liabilities,  
    ROUND(SUM(Net_Assets), 2) AS net_assets
```

FROM FMCG_FIN_OLTP.dbo.financials_fmcbg

GROUP BY

Symbol,

Company_Name,

YEAR(Period),

Fiscal_Quarter;

“

Query Explanation

- Summation logic is used to aggregate the assets and liabilities.
- Facilitates the liquidity and balance sheet analysis.
- The shareholder equity is embodied by the net assets.

Output

	symbol	company_name	year	quarter	current_assets	non_current_assets	current_liabilities	non_current_liabilities	net_assets
1	GOLG	Colgate-Palmolive (Pakistan) Limited	2020	Q2	19368.00	5960.00	1131.00	12524.00	18673.00
2	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q1	22233.00	7270.00	1778.00	11995.00	19982.00
3	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q2	22959.00	7534.00	1672.00	13757.00	21429.00
4	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q3	20424.00	6437.00	1468.00	12145.00	18648.00
5	GOLG	Colgate-Palmolive (Pakistan) Limited	2021	Q4	21305.00	6834.00	1391.00	10071.00	20013.00
6	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q1	28759.00	7816.00	1831.00	14318.00	22227.00
7	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q2	30577.00	7812.00	1134.00	13555.00	24727.00
8	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q3	24670.00	7652.00	1923.00	13299.00	21343.00
9	GOLG	Colgate-Palmolive (Pakistan) Limited	2022	Q4	25279.00	7726.00	1830.00	11359.00	22548.00
10	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q1	44226.00	8133.00	3229.00	19130.00	28017.00
11	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q2	45932.00	8336.00	3879.00	18679.00	30474.00
12	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q3	32976.00	8001.00	2818.00	13229.00	26029.00
13	GOLG	Colgate-Palmolive (Pakistan) Limited	2023	Q4	39505.00	7942.00	3064.00	17675.00	24317.00
14	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q1	51823.00	9673.00	3537.00	27380.00	31618.00
15	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q2	47411.00	9980.00	3290.00	17917.00	36211.00
16	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q3	49497.00	8665.00	19688.00	11905.00	29739.00
17	GOLG	Colgate-Palmolive (Pakistan) Limited	2024	Q4	47610.00	9052.00	3580.00	21282.00	34890.00
18	GOLG	Colgate-Palmolive (Pakistan) Limited	2025	Q3	43035.00	10048.00	3334.00	16899.00	32849.00
19	MUREB	Murree Brewery Company Limited	2020	Q2	5973.00	6393.00	959.32	10328.00	10284.00
20	MUREB	Murree Brewery Company Limited	2021	Q1	6881.00	7325.00	1110.00	10889.00	11667.00
21	MUREB	Murree Brewery Company Limited	2021	Q2	6854.00	7264.00	1100.00	10917.00	11891.00
22	MUREB	Murree Brewery Company Limited	2021	Q3	6399.00	6309.00	1202.00	10021.00	10387.00
23	MUREB	Murree Brewery Company Limited	2021	Q4	6624.00	7419.00	260.39	10688.00	11506.00
24	MUREB	Murree Brewery Company Limited	2022	Q1	0.00	0.00	0.00	0.00	12183.00
25	MUREB	Murree Brewery Company Limited	2022	Q2	6561.00	8088.00	1050.00	11258.00	12308.00
26	MUREB	Murree Brewery Company Limited	2022	Q3	7450.00	7201.00	1398.00	11782.00	12069.00
27	MUREB	Murree Brewery Company Limited	2022	Q4	7256.00	7844.00	498.56	10945.00	12183.00
28	MUREB	Murree Brewery Company Limited	2023	Q1	8814.00	8337.00	1695.00	11792.00	13786.00
29	MUREB	Murree Brewery Company Limited	2023	Q2	8924.00	8326.00	1550.00	10999.00	14141.00
30	MUREB	Murree Brewery Company Limited	2023	Q3	7483.00	7911.00	1440.00	11353.00	12741.00
31	MUREB	Murree Brewery Company Limited	2023	Q4	8257.00	8394.00	632.97	11237.00	13334.00
32	MUREB	Murree Brewery Company Limited	2024	Q1	11225.00	8542.00	1583.00	2254.00	15630.00
33	MUREB	Murree Brewery Company Limited	2024	Q2	12185.00	8520.00	1645.00	2272.00	16421.00
34	MUREB	Murree Brewery Company Limited	2024	Q3	9824.00	8354.00	2038.00	11543.00	14557.00
35	MUREB	Murree Brewery Company Limited	2024	Q4	11220.00	8610.00	511.13	15020.00	15131.00
36	MUREB	Murree Brewery Company Limited	2025	Q3	13255.00	8346.00	2111.00	2277.00	16713.00
37	NATF	National Foods Limited	2020	Q2	8956.00	6921.00	4046.00	5488.00	6344.00

7. PHASE – 3 (Power BI Implementation and DAX-Based Financial Analysis)

Introduction to Power BI Stage.

Microsoft power BI became the business intelligence and reporting layer of the project after successful development of OLAP data marts in SQL server. Power BI allows exploring data dynamically and filtering it using interactive or dynamic filters and performing more complex analytical calculations with Data Analysis Expressions (DAX). This stage had the main purpose to convert aggregated financial information into valuable conclusions using clear indicators and easy to understand visualizations.

Data Import and Model Design

Three OLAP data marts were loaded in the FMCGFINOLAP database into the Power BI:

- FinancialPerformanceMart
- DividendPayoutMart
- FinancialPositionMart

Tables indicate each quarterly financial picture at the company level. The data model was deliberately held down and kept simple and denormalized to make it highly performing and not complex to analyze.

Key Modeling Decisions

At all tables there was a preservation of grain quarterly.

Every financial number was aggregate in SQL Server.

Analytical calculations and visualization were the areas where Power BI was mostly applied.

The time-based analysis was done with Year and Quarter columns without the creation of needless complexity.

Formulas

“”””

DAX Financial Performance Analysis.

The Financial Performance Mart assists in the income statement analysis. Measures of DAX were developed to compute totals and averages which are responsive to user selections of company, year and quarter.

Core Financial Measures

Total Sales

Total Sales := SUM (FinancialPerformanceMart[sales])

Represents total revenue generated in the selected period.

Total Gross Profit

Total Gross Profit := SUM (FinancialPerformanceMart[gross_profit])

Measures profitability after direct costs.

Total Operating Profit

Total Operating Profit := SUM (FinancialPerformanceMart[operating_profit])

Indicates operational efficiency.

Profit After Tax (PAT)

Profit After Tax := SUM (FinancialPerformanceMart[pat])

Shows final profitability after taxes.

Earnings and Margin Measures

Average Earnings Per Share (EPS)

Average EPS := AVERAGE (FinancialPerformanceMart[eps])

Reflects shareholder earnings.

Average Gross Profit Margin

Average GP Margin := AVERAGE (FinancialPerformanceMart[gp_margin])

Evaluates cost efficiency.

Average Operating Profit Margin

Average OP Margin := AVERAGE (FinancialPerformanceMart[op_margin])

Indicates operational strength.

Dividend Policy Analysis Using DAX

The Dividend Payout Mart enables analysis of dividend distribution trends and shareholder return strategies.

Dividend Measures

Average Dividend Per Share (DPS)

Average DPS := AVERAGE (DividendPayoutMart[dps])

Measures dividend consistency.

Average Dividend Payout Ratio

Average Payout Ratio := AVERAGE (DividendPayoutMart[payout_ratio])

Indicates proportion of earnings distributed as dividends.

Average Net Assets Per Share

Average Net Assets Per Share := AVERAGE (DividendPayoutMart[net_assets_per_share])

Represents shareholder equity value.

Financial Position and Liquidity Analysis Using DAX

The Financial Position Mart supports balance sheet analysis and liquidity assessment.

Asset and Liability Measures

Total Current Assets

Total Current Assets := SUM (FinancialPositionMart[current_assets])

Total Non-Current Assets

Total Non-Current Assets := SUM (FinancialPositionMart[non_current_assets])

Total Assets

Total Assets := [Total Current Assets] + [Total Non-Current Assets]

Liability and Equity Measures

Total Current Liabilities

Total Current Liabilities := SUM (FinancialPositionMart[current_liabilities])

Total Non-Current Liabilities

Total Non-Current Liabilities := SUM (FinancialPositionMart[non_current_liabilities])

Total Liabilities

Total Liabilities := [Total Current Liabilities] + [Total Non-Current Liabilities]

Net Assets (Equity)

Net Assets := SUM (FinancialPositionMart[net_assets])

Key Financial Ratios

Working Capital

Working Capital := [Total Current Assets] - [Total Current Liabilities]

Measures short-term financial health.

Current Ratio

Current Ratio := DIVIDE ([Total Current Assets], [Total Current Liabilities])

Evaluates liquidity position.

Debt to Assets Ratio

Debt to Assets Ratio := DIVIDE ([Total Liabilities], [Total Assets])

Assesses solvency risk.

Equity Ratio

Equity Ratio := DIVIDE ([Net Assets], [Total Assets])

Indicates financial stability.

“””

Dashboard Design and Visualization Strategy.

The dashboard was also created so as to make the complex financial information clear, structured and decision-oriented. The focus was made on the choice of visualizations that convey the trends, comparisons, and financial connections. The dashboard pages were arranged by each one of the analytical themes, which included a financial performance, dividend policy, and financial position, to provide logical flow and easy interpretation. KPI cards, line charts, bar charts, and gauges are visuals that have been chosen on the basis of being suitable in the analysis of financial data and executive reporting.

Formatting and Interactivity.

The Power BI report was formatted consistently and interactivity to make it easier to read and use. All the financial data were standardized to two decimal points to ensure numerical accuracy and consistency in reporting, and the ratio based measures were presented as a percentage. Company, year and quarter interactive slicers were also added so that the users are able to filter and explore the data dynamically. The color scheme and the structure of the layout were applied across all of the report pages to ensure visual consistency and less cognitive load.

Power BI Solution Analytical Value.

The deployed Power BI solution is manifested by the high analytical utility by allowing one to estimate financial performance, dividend dynamics, and balance sheet robustness among various FMCG businesses and periods. OLAP data marts with their pre-aggregated data and dynamic DAX measures can be used together to generate an insight quickly without reducing the quality of data. With the help of it, users have the opportunity to recognize the trends of profitability quickly, determine the position of liquidity and solvency, and compare companies according to the main financial data, thus facilitating informed financial and strategic decision-making.

Summary of Power BI Phase

To conclude, the Power BI stage has managed to convert structured financial information to a reporting solution that is insightful and interactive. With the help of the efficient application of DAX measures, the optimized model of data and the properly defined visualizations, the system provides a complete picture of the FMCG financial performance. This step shows the real implementation of the concepts of business intelligence and shows the role of Power BI as an effective tool of financial analysis and executive reporting.

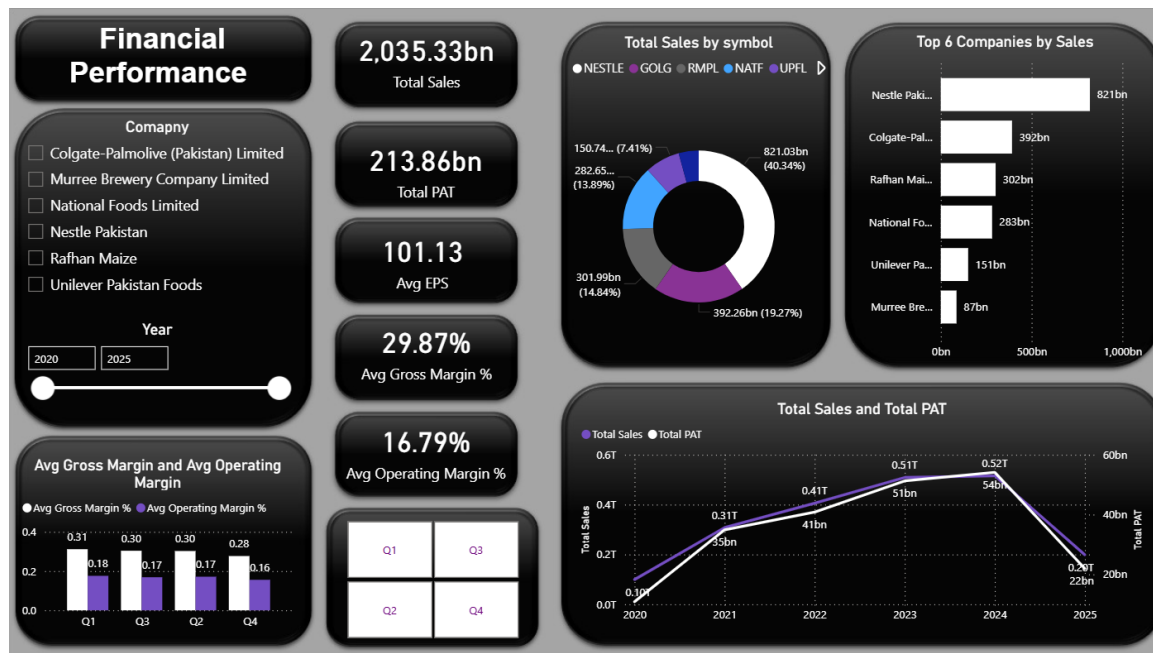
8. PHASE – 4 (Analytical Layer and Dashboard-Based Insights in Power BI)

The last phase of the project, which followed the completion of ETL process and the creation of data marts based on the OLAP, was the development of the interactive analytical system with the help of Microsoft Power BI. This layer was aimed at converting structured financial and dividend data into valuable insights that might be useful in investment assessment and financial

interpretation. It was designed with three dashboards which had correspondences to the three data marts of the project: Financial Performance, Dividend Analysis, and Financial Position.

Such dashboards are dynamically linked depending on shared dimensions like company, year and quarter that gives the user the option of filtering and analyzing data in a variety of ways. The design also allows the comparison of firms, observation of trends through time, as well as defining financial strengths and weaknesses across the FMCG industry.

Financial Performance Dashboard.



The Financial Performance dashboard is a financial performance dashboard, and the financial performance mart is based on the sales growth, profitability, and operating efficiency. On the portfolio level, the dashboard reports a total sales of about PKR 2,035 billion and profit after tax of PKR 213.86 billion. Such values denote that the FMCG companies involved in the analysis have a very large scale of operations and high profits as the consumer goods are crucial in the economy of Pakistan.

The margin indicators also support this conclusion. A gross profit margin of approximately 30 percent and operating margin of approximately 17 percent is an indication that FMCG companies can establish high pricing power without having to raise the operating cost high. The average earnings per share are also high, which proves that these companies generate high shareholder value.

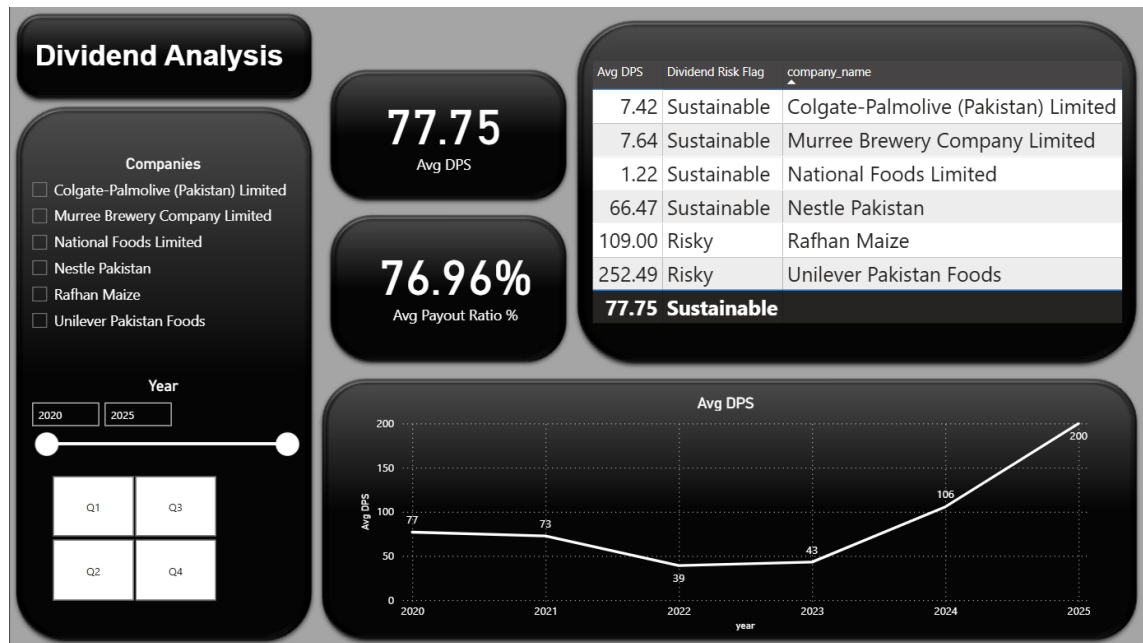
The market is highly concentrated as indicated by the sales distribution images. The market is dominated by Nestle Pakistan, which has a dominant share in the sector, with Colgate, and

Unilever and Rafhan holding the remaining share of over fifty percent. This level of concentration can be used to explain the stability in margins and profits over the years since only a few strong firms can control the prices and distribution.

Among the main lessons that this dashboard taught me are:

- Sales and profits grew continuously between the years 2020 through 2024, which means that sales and profits experienced robust post-pandemic recovery.
- The sales are closely followed by profit after tax implying good quality earnings.
- The downward trend in 2025 is probably because of half-year statistics, and not a down turn.
- On the whole, the Financial Performance dashboard proves that the chosen FMCG companies are profitable and efficient in what they do.

Dividend Analysis Dashboard.



The Dividend Analysis dashboard is created on the basis of the Dividend Payout Mart and it aims at the sustainability and credibility of shareholder returns. The dividend per share and the payout ratio of the companies of my choice are around PKR 77.75 and 77 percent respectively. This indicates that FMCG companies allocate significant percentage of their earnings to their shareholders, just like the mature, cash generating sectors.

But high payout ratio is also risky when the earnings would decrease. To this, the dashboard sorts companies as sustainable and risky dividend payers. The Company of Nestle, Colgate, Murree Brewery and National Foods are among the firms that are categorized as sustainable

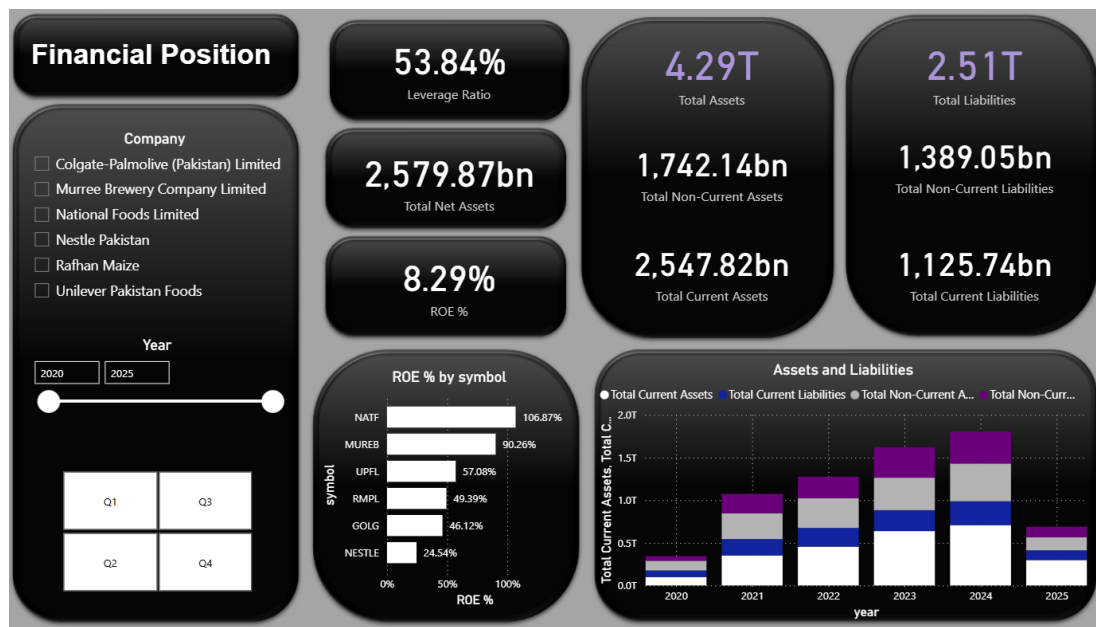
since their earnings support their dividends adequately. By contrast, Rafhan Maize and Unilever Pakistan Foods are termed as risky as they allocate a large percentage of the profits.

The dividend trend chart indicates that dividends have fallen in 2022 at the time of economic stress but they have gone up sharply since 2023. This means that dividends in the FMCG industry are highly linked with profitability as opposed to being supported by borrowing.

The most important dividend-related insights are:

- The firms in the FMCG tend to be high dividend payers.
- High dividends are not always safe and payout ratios should be taken into consideration.
- The post 2022 recovery in dividend represents the strength of earnings.

Financial Position Dashboard.



The financial position dashboard measures strength of balance sheet with the Financial Position Mart. The total assets of the selected firms amount to more than PKR 4.29 trillion, and total liabilities amount to about PKR 2.51 trillion. This yields a leverage ratio of approximately 54 percent, which means that it uses moderate amount of debt and financial risk is low.

The 8.3 percent returns on equity indicate that these companies are able to produce stable but not aggressive returns on shareholders. There are firms which have higher ROE like National Foods and Murree Brewery and have better utilization of its capital whereas larger companies like Nestle also demonstrate lower ROE because of the high number of assets that it holds.

The trend of assets and liabilities also reveal that the assets are always higher than the liabilities and are steadily increasing with the time which means a financial stability and profit re-investment.

The key findings of this dashboard are:

- FMCG companies have good equity positions.
- The levels of debts are not excessive.

Balance sheets have been fortified over the years.

Overall Interpretation

By looking at the three dashboards as a unified entity, there is a pattern of financial image. FMCG industry within Pakistan has been depicted to be lucrative in terms of profitability, financial stability and to a large extent dominated by a few robust organizations. These groups do not only make high sales and profits but they have the financial strength to pay dividend to the shareholders without a substantial increase in financial risk.

The Power BI analytical layer thus gives powerful empirical evidence in regard to the previous company choice and dividend segment in the project. The companies that are mentioned as the dividend leaders can also be noticed as financially powerful and functioning efficiently, which proves the credibility of the data and the analysis system.